

# ABHISHEK TRIPATHI

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## PROFESSIONAL SUMMARY

Associate Data Scientist with 1.5+ years of experience in machine learning, predictive analytics, and AI solution development. Published researcher in deep learning-based computer vision. Proficient in Python, SQL, and R with expertise in EDA, statistical modeling, hypothesis testing, data visualization (Tableau, Power BI), and deploying scalable ML solutions on AWS and Azure. Proven ability to translate business questions into analytical frameworks and communicate actionable insights to cross-functional stakeholders.

## TECHNICAL SKILLS

**Languages & Analytics:** Python, SQL, R

**ML & Statistics:** Hypothesis Testing, EDA, Regression, Classification, XGBoost, Neural Networks, NLP, Deep Learning (YOLOv8/v9), A/B Testing, Predictive Modeling, Forecasting, Recommendation Systems

**GenAI & LLMs:** GPT-4o, Azure OpenAI, Gemini Pro, LangChain, LangGraph, RAG Pipelines, Prompt Engineering, AI Agents

**Visualization & BI:** Tableau, Power BI, Matplotlib, Seaborn

**Cloud & Databases:** AWS (S3, ECR, Bedrock, App Runner), Azure (OpenAI, AI Search, Doc Intelligence), GCP/Vertex AI, Snowflake (familiar), MySQL, PostgreSQL, ChromaDB, FAISS, Pinecone

**Frameworks & Tools:** FastAPI, Streamlit, Docker, python-pptx, PaddleOCR, TesseractOCR, Git

## PROFESSIONAL EXPERIENCE

**Associate Data Scientist** | AlgoAnalytics Pvt Ltd, Pune, India

Jun 2025 – Present

### AlgoCreator — RAG-Powered Presentation Generator

- Architected an end-to-end GenAI product auto-generating branded PPTX decks from historical slide repositories using RAG + GPT-4o, reducing manual presentation effort by ~70%.
- Built a multi-modal extraction pipeline (python-pptx) for text, tables, and images across 1,000+ slide repositories; vectorized corpus into ChromaDB with LangChain-orchestrated generation; conducted EDA to improve retrieval accuracy by 35%.

### OneStopAI — Multi-Domain AI Platform (Team Lead)

- Led a team of 4–5 interns through full ML lifecycle: problem framing, EDA, model selection, training, validation, and deployment across manufacturing and finance verticals.
- Developed Customer Churn & Retention analytics app: applied statistical hypothesis testing, benchmarked Logistic Regression, XGBoost, and ensemble methods; extended to a personalized recommendation engine; communicated findings via Tableau and Power BI dashboards.

### Enterprise Client Engagements

- Architected an AI Recruiting Platform (Azure OpenAI + AI Search): automated JD-to-candidate matching, 24/7 AI avatar interviews, and post-interview scorecards — reducing recruiter screening time by 60%.
- Delivered production RAG-based document intelligence chatbots for enterprise clients using AWS Bedrock and Azure AI services; queried, cleaned, validated, and transformed multi-source structured and unstructured datasets at scale.

**Data Science Consultant** | AlgoAnalytics Pvt Ltd, Pune, India

Jan 2025 – May 2025

- Built an automated extraction pipeline for Japanese-language .docx/.xlsx files using PaddleOCR, LangChain, and GPT-4o; deployed as a FastAPI microservice on AWS App Runner processing 500+ documents per batch.
- Developed intelligent slide retrieval across thousands of presentations using ChromaDB multi-query semantic search with PDF/PPTX export, cutting search time from hours to seconds; maintained reproducible workflows with Git version control and documentation.

**Assistant Data Scientist** | MIT-WPU, Pune, India

Mar 2024 – Jul 2024

- Trained and benchmarked YOLOv8n and YOLOv9c deep learning models for real-time plant disease detection in hydroponics environments; conducted rigorous statistical evaluation (precision, recall, F1, mAP50, mAP50-95) — achieving 95.93% precision and 88.38% mAP50-95 with YOLOv9.

- Converted YOLOv8 model to TensorFlow Lite and contributed to Android production deployment used in live research trials; collaborated cross-functionally with domain scientists and engineers to translate domain questions into ML solutions.
- Published peer-reviewed research paper on findings: "Comparative Analysis of YOLOv8 and YOLOv9 Models for Real-Time Plant Disease Detection in Hydroponics" — Engineering, Technology & Applied Science Research (ETASR), Vol. 14, No. 5, 2024, pp. 17269–17275. DOI: 10.48084/etasr.8301.

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**PUBLICATION**

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**Tripathi, A., Gohokar, V., & Kute, R.** (2024). *Comparative Analysis of YOLOv8 and YOLOv9 Models for Real-Time Plant Disease Detection in Hydroponics*. Engineering, Technology & Applied Science Research (ETASR), **14**(5), 17269–17275. DOI: 10.48084/etasr.8301

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**EDUCATION**

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| <b>PG Diploma in AI &amp; Machine Learning</b>   MIT-WPU, Pune    | Aug 2023 – Jun 2024 |
| CGPA: 9.0 / 10                                                    |                     |
| <b>B.Sc. Mathematics</b>   CSJMU, Kanpur                          | 2020 – 2023         |
| Foundation in probability, statistics, and mathematical modeling. |                     |

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**CERTIFICATIONS**

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Introduction to AI & ML — Google Cloud | Introduction to Generative AI — Google Cloud | Data Analytics Essentials — Cisco | Python & SQL — HackerRank | Foundations: Data, Data, Everywhere — Coursera